

GSSG Pyrolysis with Fracture-Cycle Self-Healing

Test 1 Scope Specification

Independent Laboratory Engagement Brief

June 2026 · CCAL v2.1

1. Purpose

This document scopes the first of three falsification tests specified in ERES-THREE-FALSIFICATION-TESTS (JAS, 2026, CCAL v2.1). It is intended for presentation to an independent laboratory, a funding partner, or both. The test has a binary outcome: the GSSG self-healing claim either survives an adversarial pass or is withdrawn and the protocol is revised. The result is a peer-reviewable paper either way.

This scope specification is derived directly from the canonical falsification test document. Nothing here supersedes or enlarges the pre-registered criteria set out in that document.

2. Background: What GSSG Is

GSSG (Graphene-Steel-Silicon-Glass composite) is the material spine of the ERES physical architecture. It appears in three roles within the corpus:

- Test 1 (this document): laboratory specimen under Raman-spectroscopic verification of self-healing across fracture-anneal cycles
- Test 3: structural panel in a Tiny House On Wheels (THOW) across one full seasonal cycle
- Long-term: panel system for FDRV (Federated Distributed Residential Vehicle) and, at maximum scale, the interstellar vessel described by the VLSA Principle

The VLSA Principle states: if it works in a THOW, it works on a generation ship. Test 1 is the gate that must pass before Test 3 can begin, and before any commercial deployment of GSSG panels is discussed.

GSSG is also the physical link between EAAP (the attestation protocol layer) and the BERA instrumentation suite (Test 2). EAAP attests to supply-chain and construction events; BERA measures resonance within the structures that GSSG panels constitute. The three tests are therefore structurally interdependent, not parallel.

3. Test 1 Specification (Canonical)

The following is reproduced verbatim from the canonical falsification test document:

"One independent laboratory, not affiliated with the ERES Institute, replicates the GSSG pyrolysis protocol end-to-end from published parameters and subjects the resulting specimen to a pre-registered fracture-cycling regimen — minimum fifty controlled microcrack-then-anneal cycles, with load and thermal envelopes fixed in advance. The success criterion is Raman-spectroscopic confirmation of graphene reformation across cycles: the D/G band-intensity ratio must return to within a pre-registered tolerance of the pre-fracture baseline after each anneal, and the 2D-band must remain consistent with multilayer graphene structure. Falsification: if D/G drift exceeds tolerance over three consecutive cycles, or if the 2D-band collapses, the self-healing claim is withdrawn at its current strength and the protocol revised. The cost envelope to reach a definitive yes/no on this test is modest by materials-science standards — a single instrument-week on a calibrated Raman setup plus the cycling rig — and the result is a paper either way."

Source: ERES-THREE-FALSIFICATION-TESTS, JAS, 2026, CCAL v2.1.

4. Pre-Registration Requirements

Before the laboratory begins work, the following parameters must be fixed in writing and deposited to Zenodo under CCAL v2.1:

Parameter	Pre-Registration Requirement
D/G band tolerance	Numeric threshold (e.g., ± 0.05) set before first cycle runs
Load envelope	Stress range (MPa) and cycle profile for microcrack induction
Thermal envelope	Anneal temperature, ramp rate, hold time, and cooling rate
Minimum cycle count	50 cycles minimum; results reported at cycle 10, 25, and 50
Raman setup specification	Instrument model, calibration date, excitation wavelength, resolution
2D-band criterion	Qualitative consistency with multilayer graphene structure, specified by reference spectrum
Falsification trigger	D/G drift exceeding tolerance for three consecutive cycles, OR 2D-band collapse at any cycle

Pre-registration is non-negotiable. No post-hoc adjustment of tolerance or criteria is permitted after cycles begin. This is the difference between a falsification test and a demonstration.

5. Laboratory Qualifications

The laboratory engaged for Test 1 must satisfy all of the following:

- Independent: no affiliation with the ERES Institute or with any party holding a commercial interest in GSSG

- Raman-capable: calibrated Raman spectrometer with documented excitation wavelength and resolution appropriate to D/G and 2D-band discrimination in multilayer graphene
- Cycling-capable: controlled microcrack induction apparatus with programmable load and thermal envelopes meeting the pre-registered parameters
- Documented: ISO 17025 accreditation preferred; institutional affiliation with a recognised university materials science department acceptable
- Willing to co-author: the laboratory PI or lead technician must be named as co-author on the resulting paper and must agree to CCAL v2.1 licensing of the data

6. Cost Envelope

The following is a realistic all-in estimate for Test 1 as a standalone funded engagement. These are ranges, not fixed prices; the actual figures depend on laboratory jurisdiction, equipment availability, and sample preparation complexity.

Line Item	Estimated Range (USD)	Notes
Raman instrument-week	\$1,500 – \$5,000	University core facility or commercial lab
Cycling rig setup and 50-cycle run	\$2,000 – \$8,000	Lab-owned or rented apparatus
Sample preparation and materials	\$1,000 – \$4,000	GSSG specimen fabrication from protocol
Technician time and data analysis	\$3,000 – \$10,000	Lab PI, postdoc, or senior technician
Pre-registration and data deposit	\$500 – \$2,000	Zenodo deposit, DOI registration
Paper preparation and submission	\$500 – \$3,000	APC or open-access fee if applicable
Total Realistic Range	\$8,500 – \$32,000	Conservative to comprehensive

A funded engagement of \$20,000 covers the full realistic range with contingency. This is the appropriate ask for a first-milestone pilot partnership. It is modest by materials-science standards for a test that produces a peer-reviewable paper regardless of outcome.

7. Deliverables

Test 1 produces the following, regardless of outcome:

- Pre-registration deposit to Zenodo with DOI, prior to first cycle (CCAL v2.1)
 - Cycle-by-cycle Raman spectroscopy data, deposited to Zenodo upon completion
 - Pass/fail determination against pre-registered criteria, issued as a signed laboratory report
 - Co-authored peer-reviewable paper submitted to an appropriate materials science journal
 - If pass: a clean gate for Test 3 (THOW seasonal cycle) and the GSSG commercial manufacturing conversation
 - If fail: a revised GSSG pyrolysis protocol and a documented falsification record — which is itself a contribution to the field
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8. Governance and Pre-Conditions

The following pre-conditions must be satisfied before Test 1 can be funded and initiated:

- The GSSG pyrolysis protocol must be published in sufficient detail for independent replication, deposited to Zenodo under CCAL v2.1
- EMA (Emanuel M. Alexiou, Chief Social Engineer and Kingmaker) must be at the table as a co-signatory on the funding engagement — the canonical falsification test document addresses him directly for this reason
- The funding partner must accept CCAL v2.1 licensing terms for all data and the resulting paper
- The laboratory must agree to pre-registration of all parameters before cycles begin

The Kingmaker role is structural, not ceremonial. The canonical falsification test document states: “a Kingmaker chooses what may be tested, not what may be assumed.” EMA’s co-signature is the governance mechanism that distinguishes this from a demonstration.

9. Relationship to EAAP and the Broader Architecture

Test 1 is the physical gate for the entire ERES materials architecture. Its relationship to the commercial EAAP layer is as follows:

- EAAP (biometric attestation protocol) attests to supply-chain provenance, construction events, and panel installation records. It is the governance layer that rides on top of GSSG infrastructure.
- Test 1 validates the substrate that EAAP will govern. A failed Test 1 does not invalidate EAAP; it narrows the scope of what EAAP attests to until the protocol is revised.
- A passed Test 1 opens the commercial manufacturing conversation and enables the THOW pilot (Test 3), which is the VLSA Principle in its first physical instance.

For a commercial partner considering the ERES architecture, Test 1 is the earliest and cheapest way to establish whether the physical substrate of the long-term architecture is real. The ask is modest; the information value is high.

10. Contact and Attribution

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